

Q1. RIP - Chennai HQ Campus

a) Understanding of Case

In early February 2024, the Delivery IT Lead requested the addition of a temporary VLAN gateway chain for a client onboarding project in the Chennai HQ. This change (CHG-0221) was approved and implemented. Four days later, a new Delivery subnet was provisioned behind these added VLAN gateways (CHG-0224). Because the Chennai HQ network runs on the RIP v2 protocol, this chain of gateways pushed the new subnet to 16 router hops away from the core switch. The issue went completely undetected by standard monitoring systems because RIP does not generate explicit error logs for unreachable networks; it simply omits them. The NOC only discovered the routing failure when end-users filed support tickets reporting access failures at 09:42 IST on March 18.

b) Application of Concepts

RIP v2 is a distance-vector routing protocol that relies entirely on a hop-count metric to determine the best path to a destination network. The protocol enforces a strict maximum limit of 15 hops to prevent routing loops. If a network is 16 hops away or more, RIP assigns it a metric of 16. A metric of 16 represents an "infinite" distance, meaning the route is considered technically unreachable. Since the new Delivery subnet was exactly 16 hops away, the RIP process on the core switch silently discarded the route, excluding it from the routing table entirely.

c) Analysis

This fault produced no automatic alert because distance-vector protocols like RIP are designed with "silent route withdrawal" for unreachable destinations. When a metric hits 16, RIP does not view this as a configuration error to be flagged; it is simply functioning as designed by ignoring an unreachable path. This reveals a critical structural limitation: distance-vector protocols lack contextual awareness of topology health beyond simple hop counts. This limitation directly connects to the RCA finding that the organization's monitoring tools were only looking for physical layer link-down events, blinding the NOC to logical, silent routing anomalies.

d) Solution Design

Solution: The immediate technical fix is to redesign the VLSM/subnetting structure at the Chennai campus to flatten the network hierarchy, ensuring no subnet is more than 2 or 3 hops away from the core, thereby easily staying under RIP's 15-hop limit. Alternatively, migrating the campus off RIP to a protocol without hop-count limitations, such as OSPF, is recommended.

Justification: RIP v2 was originally a highly reasonable choice for the Chennai HQ campus because the environment was described as a "small, stable topology". RIP requires minimal configuration overhead, utilizes low memory, and is perfectly adequate for flat, departmental subnets (Finance, HR, Delivery) that rarely change.

e) Clarity: Before / After Route Flow

- **Before (Incident State - Failed):**
Core Switch -> VLAN 1 -> VLAN 2 -> ... -> VLAN 15 -> (Drop: Metric 16) -x- Delivery Subnet
- **After (Redesign State - Success):**
Core Switch -> Flattened Campus Core Routing -> Delivery Subnet (Metric: 2 hops)

Q2. IGRP - Bengaluru-Pune-Hyderabad Regional Link

a) Understanding of Case

An ISP-side fault occurred on the shared fiber segment between Bengaluru and Pune, which operated on a 2 Mbps link. This fault degraded the line, significantly reducing its effective bandwidth and increasing packet delay. An alternate, healthier path was available to route traffic via Pune-Hyderabad-Bengaluru. However, because the regional network runs on IGRP—a legacy distance-vector protocol—it took an unusually long time to recalculate the degraded metrics and reconverge. Consequently, traffic remained on the faulty direct path, causing severe latency (600ms+) for Bengaluru-based client applications.

b) Application of Concepts

IGRP differs from RIP by calculating a composite metric rather than relying on hop counts. It evaluates both bandwidth and delay to make routing decisions.

The standard IGRP metric formula for active parameters (\$K_1\$ and \$K_3\$) is expressed as:

$$\text{Metric} = \left(\frac{10,000,000}{\text{Bandwidth}_{\text{Kbps}}} \right) + \left(\frac{\text{Delay}_{\text{microseconds}}}{10} \right)$$

Under normal conditions, the 2 Mbps direct link to Pune likely produced a lower composite metric than traversing the two separate links via Hyderabad (8 Mbps link). When the fault occurred, the effective bandwidth dropped and the delay spiked. This caused the composite metric of the direct path to exceed the metric of the alternate Pune-Hyderabad-Bengaluru path. IGRP should have immediately switched preference to the alternate path upon calculating this new higher metric.

c) Analysis

The failure to switch paths quickly was primarily a **convergence-speed issue**, not a metric-calculation issue. IGRP successfully recalculated the composite metric, but distance-vector protocols rely on periodic updates and hold-down timers (often resulting in convergence times of several minutes). The RCA noted that this recalculation and propagation took "considerably longer than the RCA team's expectation", trapping traffic on the degraded line while the protocol slowly reconverged.

d) Solution Design

Solution: The optimal solution is to replace IGRP on the regional links with a faster-converging protocol (like OSPF or EIGRP), which is already being evaluated as part of the broader backbone rollout.

Justification: IGRP was originally expected to provide a major advantage over RIP because it uses a composite metric. This is critical for the NimbusEdge regional WAN, which features variable link speeds (e.g., a 2 Mbps link to Bengaluru and an 8 Mbps link to Hyderabad). While RIP would view a 1-hop 2 Mbps link and a 1-hop 100 Mbps link as entirely equal, IGRP intelligently prefers the higher bandwidth path. However, this advantage failed under real conditions because IGRP's slow convergence timers could not dynamically adapt to a rapid degradation event in modern enterprise networks.

Q3. OSPF Core Backbone Migration

a) Understanding of Case

On March 8, 2024, an engineer implemented change CHG-0245 to configure the Hyderabad router for OSPF Area 3. During this implementation, a transcription error resulted in the assignment of an incorrect Area ID. Because OSPF requires strict matching parameters to form neighbour relationships, the Hyderabad router never formed an adjacency with the Chennai Area 0 backbone router. Standard monitoring only verified that the physical port was up, completely failing to detect that logical routing traffic was being black-holed until trace routes confirmed the silent drops at 12:30 IST.

b) Application of Concepts

OSPF routers exchange Hello packets to discover neighbors and establish adjacencies. For two routers to move from the *Init* state to a *Full Adjacency*, several parameters within the Hello packet must match exactly:

- **Area ID**
- Hello / Dead intervals
- Authentication credentials
- Subnet Mask

Because the Hyderabad router was assigned a mismatched Area ID, the Chennai backbone router rejected its Hello packets. Consequently, no Link-State Advertisements (LSAs) were exchanged, no topology database was built, and the routers refused to inject each other's subnets into their routing tables.

c) Analysis

This fault was significantly harder to detect than a physical link failure because standard Layer 1/2 monitoring tools only observe the physical link state. The interface on the switch was technically "UP," so green lights appeared on the NOC dashboards. The root of the issue was a severe organizational gap: no formal change-freeze or validation window was scheduled during

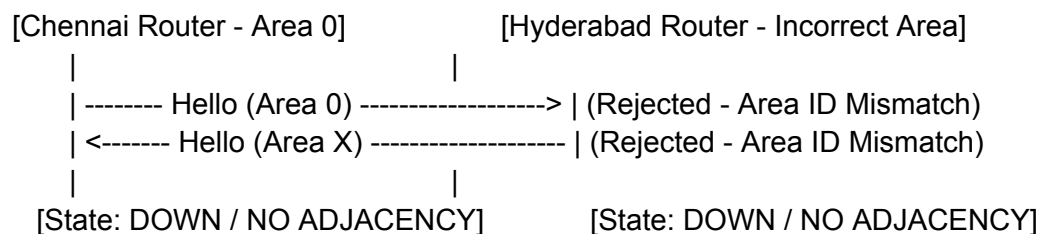
the OSPF migration. A single transcription error went live because it bypassed peer-review protocols.

d) Solution Design

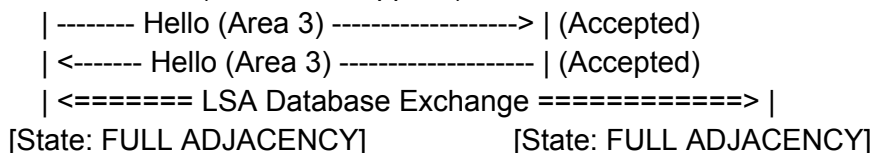
Solution: 1. **Technical Fix:** The immediate correction is to modify the OSPF process on the Hyderabad router and input the correct Area ID (Area 3) so it matches the expected Chennai interface. 2. **Process Safeguard:** Implement a mandatory adjacency verification checklist utilizing `show ip ospf neighbor` and mandate peer-reviewed Area ID assignments prior to committing changes.

Justification: NimbusEdge is migrating to multi-area OSPF because it provides immense hierarchical scalability and fast convergence. By utilizing a multi-area design (Area 0 core, Area 1-3 regional), routing updates are compartmentalized, saving memory and CPU. Furthermore, OSPF's Shortest Path First (SPF) algorithm reacts to topology changes in milliseconds rather than relying on the slow, timer-based updates seen in RIP and IGRP.

e) Clarity: OSPF Adjacency State Diagram



****Post-Fix State (CHG-0258 Applied):****



Q4. BGP ISP Peering and Failover

a) Understanding of Case

At 14:50 IST, the primary ISP (ISP-A) suffered a fiber cut, taking down NimbusEdge's leased line. The network was configured to automatically failover to the secondary ISP (ISP-B) via BGP. However, the failover did not trigger, stalling all outbound client-facing traffic. A senior network architect had to manually intervene (CHG-0259) at 15:40 IST to force the outbound traffic toward ISP-B. Due to this total loss of external connectivity, six enterprise clients were impacted for over 9 hours, resulting in a formal SLA breach of their 99.9% uptime guarantee.

b) Application of Concepts

BGP does not use bandwidth or delay; it relies on complex path attributes to determine best routes. The key attribute for determining outbound routing preference in a single Autonomous System (AS) is **LOCAL_PREF** (Local Preference). The NOC Change Log indicates that **LOCAL_PREF** values for outbound preference were modified in CHG-0249 but never re-tested end-to-end. The likely misconfiguration was that the high **LOCAL_PREF** tying outbound routes to ISP-A was not conditionally tied to reachability tracking. When the physical fiber was cut far upstream, the local interface stayed active, the BGP policy stubbornly favored ISP-A, and the AS-PATH outbound traffic was routed into a dead connection.

c) Analysis

An untested failover policy is an exponentially higher risk in BGP than in interior gateway protocols (IGPs) like OSPF. IGPs dynamically map the local topology and will automatically route around dead links. BGP, on the other hand, blindly trusts administrative policies (like **LOCAL_PREF**). If a policy tells BGP to send traffic out ISP-A, it will do so until the local session hard-drops. This single policy oversight caused a massive chain reaction: failed automatic failover -> stalled outbound traffic -> total unreachability for client applications -> failure to maintain the contracted 99.9% availability -> formal SLA breach notice issuance.

d) Solution Design

Solution: The BGP outbound policy must be corrected to include dynamic reachability tracking (such as BFD - Bidirectional Forwarding Detection, or SLA IP tracking) tied to the **LOCAL_PREF** statement. Furthermore, the organization must establish a quarterly scheduled maintenance window to physically test the BGP failover to validate AS-PATH prepending and **LOCAL_PREF** behavior.

Justification: Despite this failure, BGP is the absolute necessary choice for NimbusEdge's dual-ISP edge. Interior protocols (RIP, OSPF) cannot process the global internet routing table (hundreds of thousands of routes) and do not natively support complex inter-domain routing policies. BGP is specifically designed for Autonomous System peering, allowing NimbusEdge to maintain granular control over its external enterprise traffic.

e) Clarity: BGP Failover Logic Check

Failed BGP Logic (March 18):

- **ISP-A Route:** **LOCAL_PREF** = 200 (Static, no tracking)
- **ISP-B Route:** **LOCAL_PREF** = 100
- **Result:** ISP-A drops remotely. Local BGP session stays active. Router forces traffic to **LOCAL_PREF** 200 (ISP-A). Traffic black-holed.

Corrected BGP Logic (Proposed):

- **ISP-A Route:** **LOCAL_PREF** = 200 **TRACKING Link-State**
- **ISP-B Route:** **LOCAL_PREF** = 100

- *Result:* ISP-A drops remotely. Tracking flags the path as dead. BGP withdraws the 200 preference. ISP-B (100) becomes the highest available **LOCAL_PREF**. Traffic automatically fails over.